

Elan Hong

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Education

University of Michigan

B.S. in Data Science, B.A. in Economics

Ann Arbor, MI
Expected May 2028

Relevant Coursework: Data Structures & Algorithms, Web Systems, Data Management Systems, Data Mining, Intro to ML, Statistics & AI, Probability, Linear Algebra, Computer Organization, Discrete Math, Calculus III

Experience

AT&T

Data Science Engineering Intern

Dallas, TX
June 2026 – August 2026

- Predicted employee attrition (**0.87 ROC-AUC**) by creating an XGBoost classification model (**scikit-learn, pandas**) on **15000+** HR records, flagging the highest-risk **15%** to create 3 retention recommendations
- Cut pitch prep time by **40%** and lifted upsell conversion by **12%** by building an **agentic workflow** application that generated personalized pitches and local trends for sales associates from a **23M-account** dataset
- Cut reporting cycles by **30%** for **5+** business partners by engineering **Power BI** dashboards and **Power Query** models to track **15+** workforce KPIs on a recurring basis, replacing manual Excel workflows

Riot Games

Student ML Engineer

Seattle, WA
February 2025 – May 2025

- Compiled a CSV dataset of over **60,000** matches to **evaluate Swiftplay adoption** by collaborating with Riot Games employees to build data pipelines and scrape and store match data using breadth-first search
- Achieved **85.2%** accuracy and **0.903 AUROC** by building a **logistic regression model** using Summoner Level and performance to classify casual vs. competitive Swiftplay players and inform Ranked readiness
- Generated actionable **business insights** on player engagement trends by conducting statistical analysis and using data storytelling to communicate gameplay recommendations to non-technical stakeholders

Fantapak

Data Engineering Intern

Livonia, MI
May 2025 – August 2025

- Cut purchasing forecast errors by **20%** by training a **gradient-boosted regression model** (**XGBoost, scikit-learn**) on 12 months of historical order data to forecast monthly demand for individual SKUs
- Cut form completion time by **70%** across **500+ monthly orders** by building an automated purchase order entry application with **JavaScript, Python (Flask)**, and **SQLite** to streamline purchasing workflows

University of Michigan - Statistics Online Computational Resource Program

AI Engineer

Ann Arbor, MI
January 2026 – Present

- Enabled translation of natural-language instructions into executable **R** code and visual statistical analyses by developing an **LLM** analytics web application integrating **OpenAI GPT-4o/GAIM** capabilities
- Delivered summary statistics, linear regression models, and **interactive Plotly visualizations** for user-generated and uploaded datasets by building data analysis workflows using **R** and **Plumber REST APIs**

Projects

NYC Taxi Data Platform | Python, Kestra, DuckDB, BigQuery, dbt, Docker

- Transformed **10M+ NYC taxi trip records** into analytics-ready datasets by building a transportation data platform using **Kestra** to orchestrate automated ingestion and validation workflows into **BigQuery**
- Reduced pipeline runtime by **60%** compared with a **pandas** implementation by engineering batch **ETL** pipelines with **Python** and **DuckDB** for local querying, cleansing, and aggregation of raw Parquet data
- Reduced downstream query runtime by **40%** by developing a **dbt analytics layer** on **BigQuery** with **20+ tested dimensional models**, automated data-quality checks, and reusable SQL transformations

Research Paper RAG Assistant | Python, PyTorch, Hugging Face, OpenAI API, FAISS, LangChain

- Enabled evidence retrieval and comparison across **5,000+ research papers** by building a **RAG-based** assistant that generated cited summaries of **consensus, conflicting findings, and supporting evidence**
- Improved evidence retrieval across papers with varied terminology and writing styles by building a multi-stage **NLP pipeline** for paper parsing, semantic chunking, transformer embeddings, and **FAISS vector retrieval**

Skills & Interests

Languages : Python, C++, Java, SQL, R, JavaScript

Tools/Frameworks : Github, Tableau, Power BI, MySQL, Databricks, PostgreSQL, Excel, Hadoop, AWS, Spark, Next.js, Airflow, Docker, Kubernetes, MATLAB, Hive, Snowflake, BigQuery, dbt, FastAPI, LangChain, PyTorch

Libraries : React, Pandas, BeautifulSoup, Numpy, Matplotlib, Seaborn, Scikit-learn, Jinja, Flask, Pytorch

Clubs : Michigan Data Consulting, Product Motion, Kappa Theta Pi Tech Fraternity, Revolution Chinese Yo-yo

Interests : Chess, Chinese Yo-yo, Baking, Jazz Bars, Rock Climbing, Reading, Skiing, and Drawing